



3DEXPERIENCE®

MONTHLY PARTNER MEETING

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DS DASSAULT
SYSTEMES | The 3DEXPERIENCE® Company

安装准备

<http://pdir/?pdir=simulia,ep622,update02&context=all>

Prerequisites

This document is provided as a guideline regarding hardware configurations, operating system, and operating system patches recommended for the use with this level of Abaqus products. Consult the [Platform Support Policy](#) for definitions of support terms. For additional details on test configurations, graphics devices, performance, and selected products' compatibility matrices, go to the Dassault Systèmes Home Page (www.3ds.com) and select **Support -> Hardware and Software**.

Supported Platforms

OS	Windows 10 Professional and Enterprise	Windows Server 2016, 2019 (with or without HPC Pack)	Red Hat Enterprise Linux Server 7, 8	SuSE Linux Enterprise Server 12, 15
Processor	x86-64 Intel or AMD	x86-64 Intel or AMD	x86-64 Intel or AMD	x86-64 Intel or AMD
Required OS Components (see Notes below)			Runtime libraries as listed in Notes below	Runtime libraries as listed in Notes below

Supported Virtualization

Vendor	Product	Windows	Red Hat Enterprise Linux	SuSE Linux Enterprise Server
VMware	ESXi	✓	✓	✓
Microsoft	Hyper-V	✓	✓	✓
SUSE	XEN and KVM from SUSE Linux Enterprise Server	✗	✗	✓

Definitions of the terms related to the platform support policy are available [here](#).

Support Policy for Operating Systems

- Windows
 - The support policy on Windows 10 is available [here](#)
- Windows Server
 - Windows Server 2016 is a Qualified platform
 - Windows Server 2019 is a Validated platform
- Linux*:
 - Red Hat Enterprise Linux Server
 - Red Hat Enterprise Linux Server 7.6 is a Qualified platform
 - Red Hat Enterprise Linux Server 7.x, where $x \geq 5$, is a Compatible platform
 - Red Hat Enterprise Linux Server 8.1 is a Validated platform
 - Red Hat Enterprise Linux Server 8.x, where $x > 1$, is a Compatible platform
 - SuSE Linux Enterprise Server
 - SuSE Linux Enterprise Server 12 SP4 is a Qualified platform
 - SuSE Linux Enterprise Server 12 SPx, where $x > 4$, is a Compatible platform
 - SuSE Linux Enterprise Server 15 SP1 is a Validated platform
 - SuSE Linux Enterprise Server 15 SPx, where $x > 1$, is a Compatible platform

* Linux platform notes:

- The Abaqus CAD Associative Interfaces for CATIA v5, Pro/ENGINEER, and Solidworks are not supported on the Linux platform.
- For Haswell and newer processors a kernel update may be required. For more information, see bug number [BR10000165399](#).
- Abaqus/CAE users may see this warning on startup: "/usr/lib64/libjpeg.so.62: no version information available". This message can safely be ignored.
- THP should be disabled or set to [madvise](#). For more information, see [QA00000043172](#)

Disk Space Requirements for Installations

Product	Windows	Linux
EXALEAD CloudView	1.22G	1.46G
EXALEAD Search for Documentation	306M	301M
Isight	1.05G	1.01G
Established Products	4.01G	3.62G
Documentation	1.02G	1.02G

Support Policy for MPI and Interconnects

- MPI:
 - Windows (user-supplied):
 - MS MPI 9.0.1 is Qualified
 - Note: If MS MPI is not present on a system, 9.0.1 will be automatically installed. Newer versions of MS MPI not listed are Compatible.
 - Linux (supplied with the application):
 - IBM Platform MPI is the default MPI implementation for Abaqus/Standard
 - Intel MPI is the default MPI implementation for Abaqus/Explicit
 - Intel MPI
 - Intel® MPI 2019.7.217 is Qualified
 - Any version of Intel® MPI 2019 is Compatible
 - IBM Platform MPI
 - IBM Platform MPI 9.1.4.3 is Validated
- Interconnects:
 - Windows:
 - Infiniband: Mellanox ConnectX VPI (MT26428) - PCIe 2.0 5GT/s, IB QDR / 10GigE Network Adapter with Mellanox driver version 2.1.3.7290 is Qualified
 - GigE, 10GigE with TCP/IP Ethernet driver is Validated
 - On Windows HPC systems, any interconnect solution supported by MS MPI is Compatible
 - For more information on availability of DMP on Windows/X86-64, see [Support for Distributed Memory Parallel Analyses on Windows Platforms](#).
 - Linux:
 - Infiniband: Mellanox Technologies MT27700 Family [ConnectX-4] with OFED-4.5-1.0.1 is Qualified
 - Infiniband: Mellanox Technologies MT28908 Family [Connect-X 6] with OFED-4.9.2 is validated
 - Intel Omni-Path: Intel 100 Series with driver 10.7.0.0.145 is Qualified
 - GigE, 10GigE with TCP/IP Ethernet Driver are Validated
 - Any interconnect solution supported by IBM Platform MPI 9.1.4.3 is Compatible. See the IBM Platform MPI release notes for more information.
 - Any interconnect solution supported by Intel MPI is Compatible. See the Intel MPI release notes for more information.

Support Policy for Workload Managers

- Windows:
 - Windows HPC Server 2012 R2 Update is Validated
 - Windows HPC Server 2016 is Validated
 - All other Microsoft-supported versions of Windows HPC are Compatible
- Linux:
 - IBM Spectrum LSF 10.1.0.8 is Qualified
 - All other IBM-supported versions of Spectrum (formerly Platform) LSF are Compatible

Support Policy for FSI Co-simulation using MpCCI

- MpCCI version 4.4.2 is Qualified. For more information, see [Fraunhofer SCAI](#).

Support Policy for GPGPU Analysis Compute

GPUs with a minimum FP64 rate of 1 TFLOPs are highly recommended. GPGPU computing is not supported on VDI desktops.

- Windows
 - AMD:
 - Validated devices and drivers:
 - AMD Radeon Pro V11 with driver 20.Q1 - 19.40.01.22
 - NVIDIA:
 - Driver should be run in TCC mode for reliable GPGPU analysis computing
 - Any driver version that supports CUDA 11 or greater is Compatible (driver version 451.48 or later)
 - Validated devices and drivers:
 - Tesla K40m, GP100 with driver 452.96
- Linux
 - NVIDIA
 - Any driver version that supports CUDA 11 or greater is Compatible (driver version 450.51 or later)
 - Validated devices and drivers:
 - Tesla K40m, K80, A100, P100, V100 with drivers 450.80.02 and 460.73.01

Policy for Viewing Online Documentation

- For more information, see [HTML documentation](#).

Additional Notes

• Runtime Libraries

- The Intel® MKL 2019 Update 4 runtime math library is included with Abaqus.
- On Linux, Abaqus requires the following additional libraries that may not be installed with your distribution.
 - libjpeg 6.2 64-bit
 - libstdc++ 4.7 64-bit
 - libgomp (Red Hat/CentOS) / libgomp1 (SLES)
 - libXm4 (SLES)
 - freetype
 - ksh
 - redhat-lsb-core-4.x (Red Hat)
 - lsb-release-2.0 (SLES)
 - libpng12-0 (SLES)
 - zlib-devel (SLES)

• Memory

- A minimum of 1 GB of free physical memory is required to run Abaqus.

• Graphics

- For Abaqus/CAE and Abaqus/Viewer, X-Windows is required on Linux platforms. The OpenGL shared library is required on the computer where these products are run.
- [Graphics devices](#) that have been currently tested with 2022 GA

• Compilers:

- A C++ or FORTRAN compiler is needed for compiling user subroutines or for building post processing applications (Abaqus make utility).
- For Windows platforms only, a FORTRAN compiler must also be installed when building user subroutines. For more information, see [BR10000186112](#).
- User-written post processing programs created with the Abaqus make utility should be compiled using the compiler for the program language and linked using a C++ compiler. Therefore, both a FORTRAN and C++ compiler are required for user-written FORTRAN programs.
- User-written programs that were compiled and linked with the Abaqus make utility in a previous release of Abaqus and processing only the results (.fil) file can still be used with an Abaqus 2016 results file without any change. In this instance no compiler is required.

• Windows:

- Microsoft Visual Studio 2019 (Visual C++) is a Qualified Platform.
- Intel® Fortran Version 19.0.5 is a Qualified Platform. Versions higher than 19.0.5 are Compatible.
- Intel® Fortran on Windows requires the Microsoft Visual Studio development environment. For more information, see the [Intel Visual Fortran Pre-requisites](#).
- The Microsoft MPI Software Development Kit is required for building user subroutines that utilize MPI based parallelization. For more information, see [Microsoft MPI v9](#)

• Linux:

- Intel® Visual Fortran Version 19.0.5 is a Qualified Platform. Versions higher than 19.0.5 are Compatible.
- C++ on Linux:
 - GCC 8.2.1 is a Qualified platform
 - GCC 8.x where x>2 is a Compatible platform

• Input Devices

- Abaqus/CAE/Viewer provides support for 3DConnexion input devices such as SpacePilot, SpaceExplorer, and SpaceNavigator. Both Windows and Linux platforms are included. For best compatibility, please install the latest 3DConnexion driver and software.

Test Configurations for Abaqus 2022 Products

To view the platform configurations used to qualify Abaqus, click the tabs below. Configuration details are listed by operating system. Explanations of the table contents and terminology are provided after the tables

Windows/x86-64	
Operating System	Target Release
Windows10 Enterprise Build19042	2022
Windows10 Enterprise Build19043	2022

Linux/x86-64	
Operating System	Target Release
RedHat Enterprise Linux Server release 7.6	2022
RedHat Enterprise Linux Server release 7.9	2022
RedHat Enterprise Linux Server release 8.3	2022
RedHat Enterprise Linux Server release 8.4	2022
SLES 12	2022
SLES 15	2022

Headings for each test configuration list the operating system level and target Abaqus release. All test configurations for a particular platform are supported using the same Abaqus software build. Given the large number of possible configuration permutations, even for a single platform, SIMULIA tests only a sample of the possible configurations. Test configurations listed here represent the Abaqus configurations which have undergone, or are scheduled for, formal qualification testing. See the Program Directories -> Prerequisites -> Support Policy for Operating Systems for information on qualified, validated, and compatible platforms.

The records contain details of system configurations used during formal testing. The date listed in the individual system configuration records indicates the date the system configuration was recorded, not the date of qualification. The output includes the processor specification and the number of processors and other system parameters such as graphics details (if applicable), the amount of physical and virtual memory, and file system details.

Final configurations depend on successful testing outcomes. Supported platforms and configurations are subject to change without notice

Platform Support Policy for Virtualization

Refer to the "Platform Support Policy for Virtualization for Servers" found in the 3DX Program Directory under **Deployment -> Strategic Platforms -> Servers -> H/W & S/W Requirements**.

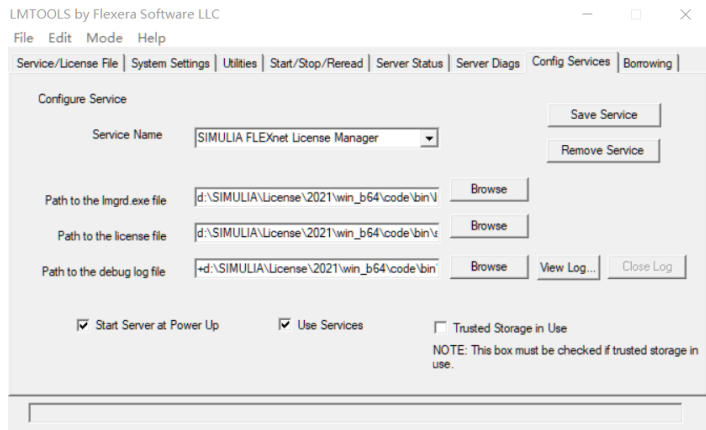
Additionally, the VDI configurations posted [here](#) are Compatible platforms for Abaqus/CAE.

FLEXNET服务器

配置

• 服务器

- Utility tool配置
- 检查、重启服务器状态



• 客户端 : *install_dir/os/SMA/site/custom_v6.env*

- *abacuslm_license_file="portnumber@licensever"*
- *license_server_type=FLEXNET*

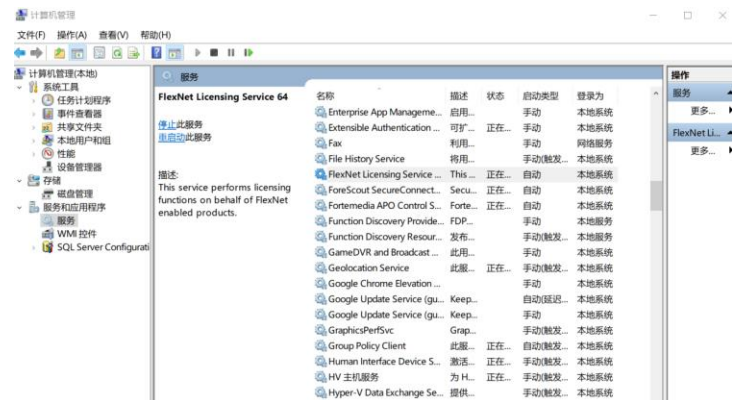
```
D:\>cd D:\SIMULIA\License\2021\win_b64\code\bin

D:\SIMULIA\License\2021\win_b64\code\bin>lmstat.exe -c simulialm.lic -a
lmstat - Copyright (c) 1989-2015 Flexera Software LLC. All Rights Reserved.
Flexible License Manager status on Fri 12/2/2022 13:08

[Detecting lmgrd processes...]
License server status: @LP15-3LI20-CHN
License file(s) on LP15-3LI20-CHN: simulialm.lic:

lmgrd is not running: Cannot connect to license server system. (-15,570)

D:\SIMULIA\License\2021\win_b64\code\bin>
```



合并LICENSE

- 不能合并

- 相同的package
- 不同的package
- 相同的feature

- 可以合并

- 相同的mac id, 两个不同的feature
- 相同的package中, 一个feature, 另一个increment

合并LICENSE

```
SERVER this_host 507b9de0b43e
VENDOR ABAQUSLM
PACKAGE abaqus ABAQUSLM 61.9 COMPONENTS="cfd explicit foundation \
simflow standard cse_token safe_ex_engine toska_topo \
toska_shape toska_bead toska_sizing toska_int_abaqus \
toska_int_ansys toska_int_nastran toska_adv_nonlinear \
toska_adv_durability toska_adv_morph toska_smooth toska_report \
tfluid_topo tfluid_smooth tfluid_parallel tfluid_int_ccmp \
tfluid_int_fluent" OPTIONS=SUITE SIGN=98790FB403CC
PACKAGE abaqus_extended ABAQUSLM 61.9 COMPONENTS="cae rtgateway \
gateway_safe_ex_gui" OPTIONS=SUITE SIGN=583506DE5BDA
FEATURE abaqus ABAQUSLM 61.9 05-dec-2018 20 VENDOR_STRING=hks \
SUPERSEDE vendor_info=IPWm5n09qbzdaa9tpAPZqa \
ISSUED=06-dec-2017 NOTICE="Chennai_DS India Priv Ltd_" \
SN=DSIND012 START=05-dec-2017 SIGN=2930351C2C26
FEATURE ams ABAQUSLM 61.9 05-dec-2018 1 VENDOR_STRING=hks SUPERSEDE \
DUP_GROUP=SITE vendor_info=CM2vm+xpRUIRVZOMlWEPg \
ISSUED=06-dec-2017 NOTICE="Chennai_DS India Priv Ltd_" \
SN=DSIND012 START=05-dec-2017 SIGN=8ECC250242D2
SN=DSIND012 START=05-dec-2017 SIGN=2648264EA7DE
FEATURE abaqus_extended ABAQUSLM 61.9 05-dec-2018 4 VENDOR_STRING=hks \
SUPERSEDE vendor_info=x11s+pk386prpxre2Rnq9g \
ISSUED=06-dec-2017 NOTICE="Chennai_DS India Priv Ltd_" \
SN=DSIND012 START=05-dec-2017 SIGN=06F50B0AA9C6
```

```
SERVER BRDTSBLRLIC4 00155d27b467
VENDOR ABAQUSLM
PACKAGE abaqus ABAQUSLM 61.8 COMPONENTS="adams cae cfd explicit \
foundation motdflow standard cse_token viewer" OPTIONS=SUITE \
SIGN=707083A48C38
FEATURE abaqus ABAQUSLM 61.8 29-jun-2018 32 VENDOR_STRING=commercial \
CAPACITY SUPERSEDE vendor_info=AobT83vvei664IktuXsuig \
ISSUED=08-nov-2017 NOTICE="LARSEN _ TOUBRO LTD" \
SN=200000000014856 START=07-nov-2017 SIGN=B9A4E0BCE344
FEATURE ams ABAQUSLM 61.8 29-jun-2018 1 VENDOR_STRING=commercial \
SUPERSEDE DUP_GROUP=SITE vendor_info=53N8vpgBk3491w6SUL1ymQ \
ISSUED=08-nov-2017 NOTICE="LARSEN _ TOUBRO LTD" \
SN=200000000014856 START=07-nov-2017 SIGN=3D682704132A
FEATURE aqua ABAQUSLM 61.8 29-jun-2018 32 VENDOR_STRING=commercial \
SUPERSEDE DUP_GROUP=SITE vendor_info=6z1P59qRqXlTvKx7X5urKg \
ISSUED=08-nov-2017 NOTICE="LARSEN _ TOUBRO LTD" \
SN=200000000014856 START=07-nov-2017 SIGN=72CAB8146302
FEATURE cosim_acusolve ABAQUSLM 61.8 29-jun-2018 1 \
VENDOR_STRING=commercial SUPERSEDE DUP_GROUP=SITE \
vendor_info=yvp5kux+fULodEdJ5XnbvA ISSUED=08-nov-2017 \
NOTICE="LARSEN _ TOUBRO LTD" SN=200000000014856 \
START=07-nov-2017 SIGN=57629FE039BA
FEATURE cosim_direct ABAQUSLM 61.8 29-jun-2018 1 \
VENDOR_STRING=commercial SUPERSEDE DUP_GROUP=SITE \
vendor_info=a13zA18isy6LVRIudFXjw ISSUED=08-nov-2017 \
NOTICE="LARSEN _ TOUBRO LTD" SN=200000000014856 \
START=07-nov-2017 SIGN=80E81388AD50
FEATURE cse ABAQUSLM 61.8 29-jun-2018 1 VENDOR_STRING=commercial \
SUPERSEDE DUP_GROUP=SITE vendor_info=B1iyqXLI9XxSBqwk35yvwg \
ISSUED=08-nov-2017 NOTICE="LARSEN _ TOUBRO LTD" \
SN=200000000014856 START=07-nov-2017 SIGN=1EF39A8E263A
FEATURE design ABAQUSLM 61.8 29-jun-2018 32 VENDOR_STRING=commercial \
SUPERSEDE DUP_GROUP=SITE vendor_info=mRN23jxyvDTrKNwMu5kygKa \
ISSUED=08-nov-2017 NOTICE="LARSEN _ TOUBRO LTD" \
SN=200000000014856 START=07-nov-2017 SIGN=1D91C600B8A6
```

不能合并

合并LICENSE

```
SERVER bng1winlic01 005056b4663f 27002
VENDOR ABAQUSLM
PACKAGE abaqus ABAQUSLM 61.9 COMPONENTS="cfd explicit foundation \
standard cse token" OPTIONS=SUITE SIGN=24BA3258EE8E
FEATURE abaqus ABAQUSLM 61.8 04-feb-2019 36 VENDOR_STRING=commercial \
SUPERSEDE vendor_info=4Ke3nsHCXOXWPz+crfXMZQ \
ISSUED=01-feb-2018 NOTICE="GE INDIA INDUSTRIAL PRIVATE \
LIMITED" SN=200000000006051 START=31-jan-2018 \
SIGN=23F80D401664
FEATURE pydriver ABAQUSLM 61.8 04-feb-2019 1 VENDOR_STRING=commercial \
SUPERSEDE DUP_GROUP=SITE vendor_info=otefujCHHh56uxwcVD7nsQ \
ISSUED=01-feb-2018 NOTICE="GE INDIA INDUSTRIAL PRIVATE \
LIMITED" SN=200000000006051 START=31-jan-2018 \
SIGN=7D9CCB4AD19C
FEATURE cae ABAQUSLM 61.8 04-feb-2019 1 VENDOR_STRING=commercial \
SUPERSEDE vendor_info=djvs0Naw8PNdbSzocheTug \
ISSUED=01-feb-2018 NOTICE="GE INDIA INDUSTRIAL PRIVATE \
LIMITED" SN=200000000006051 START=31-jan-2018 \
SIGN=721570F84950
FEATURE ams ABAQUSLM 61.9 04-feb-2019 1 VENDOR_STRING=commercial \
SUPERSEDE DUP_GROUP=SITE vendor_info=swjAMD8uEsuycxBfXLRNvg \
ISSUED=01-feb-2018 NOTICE="GE INDIA INDUSTRIAL PRIVATE \
LIMITED" SN=200000000006051 START=31-jan-2018 \
SIGN=4164CB78F6A8
FEATURE aqua ABAQUSLM 61.9 04-feb-2019 5 VENDOR_STRING=commercial \
SUPERSEDE DUP_GROUP=SITE vendor_info=0Ze0AqGcFrRFnc4wwKlCHQ \
ISSUED=01-feb-2018 NOTICE="GE INDIA INDUSTRIAL PRIVATE \
LIMITED" SN=200000000006051 START=31-jan-2018 \
SIGN=FD513B780DB2
FEATURE
```

```
SERVER bng1winlic01 005056b4663f 27002
VENDOR ABAQUSLM
INCREMENT abaqus ABAQUSLM 61.9 04-feb-2019 5 VENDOR_STRING=commercial \
SUPERSEDE vendor_info=6G0DB0mwLMERYVmbS92bgQ \
ISSUED=01-feb-2018 NOTICE="GE INDIA INDUSTRIAL PRIVATE \
LIMITED" SN=200000000006051 START=31-jan-2018 \
SIGN=600A7D6A9C2E
INCREMENT cae ABAQUSLM 61.9 04-feb-2019 1 VENDOR_STRING=commercial \
SUPERSEDE vendor_info=zha5pqwJC0E7eN+J!TRpsw \
ISSUED=01-feb-2018 NOTICE="GE INDIA INDUSTRIAL PRIVATE \
LIMITED" SN=200000000006051 START=31-jan-2018 \
SIGN=725EA34EA3DE
```

Copy the from increment and
add it to main license file

可以合并

ABAQUS LICENSE

Can configure two license file on same Machine	NO
Can combinedly use token from two different License server	NO
Can run the DSLS and Flexnet on same machine	YES
Can configure the both DSLS and Flexnet license in the client machine	NO

LICENSE服务器多个物理地址

- 如果使用第2个MAC生成license

- License 服务不能启动
- 将显示hostid不匹配

- 解决办法

- 使用第1个MAC生成license
- 改变网卡顺序

Windows IP Configuration

```
Host Name . . . . . : TCOP
Primary Dns Suffix . . . . . : corp.mahindra.com
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : corp.mahindra.com
```

Ethernet adapter Local Area Connection 7:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) B2576NS Gigabit Network Connection #2
Physical Address. . . . . : 00-19-99-CA-A5-CF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

1

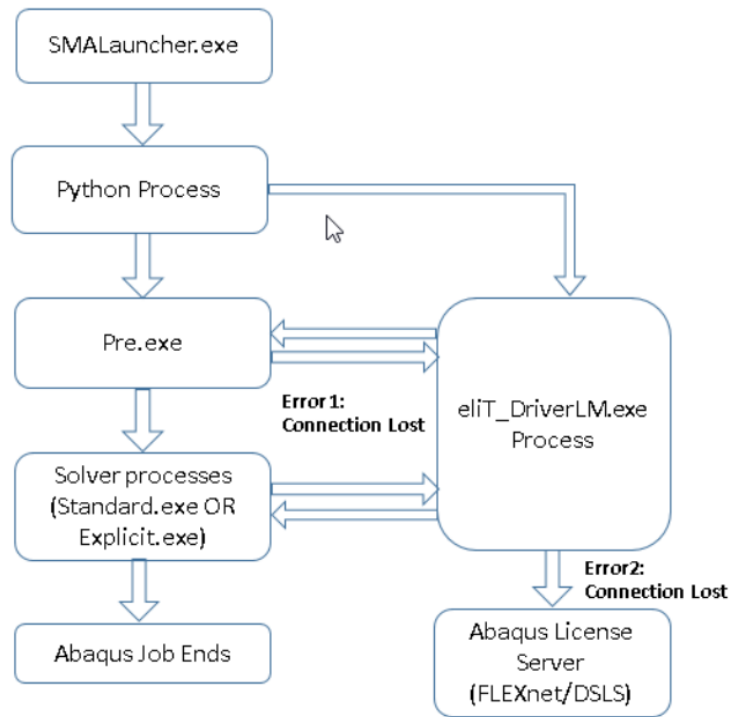
Ethernet adapter CAE_LICLAN:

```
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) B2576NS Gigabit Network Connection
Physical Address. . . . . : 00-19-99-CA-A5-CF
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::54a9:e3a8:6403:462d%30(Preferred)
IPv4 Address. . . . . : 10.210.6.160(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.210.6.1
DHCPv6 IAID . . . . . : 754981273
DHCPv6 Client DUID. . . . . : 00-01-00-01-19-47-61-D6-E4-11-5B-DE-1F-68
DNS Servers . . . . . : 10.210.0.4
NetBIOS over Tcpip. . . . . : Enabled
```

2

TOKEN没有正常返回

- Job 没有正常终止，导致某些进程仍然停留在计算节点上
 - 尝试手动杀掉进程 eliT_DriverLM.exe, pre.exe, standard.exe, packager.exe, explicit.exe
- 增加timeout机制（flexnet文件夹中abaquslm.opt文件）
 - TIMEOUT explicit 3600
 - TIMEOUT standard 3600
 - TIMEOUT abaqus 3600



客户端无法获取LICENSE

- 很大部分原因是防火墙/安全软件问题！
- 关闭客户端/服务端防火墙和杀毒软件
- 客户端ping服务器 ping server_IP
- 端口号测试
 - telnet server_ip port
- 增加ip和hostname到hosts文件中
- 发送license.log文件到技术支持团队

DSLS服务器

基本情况

- 管理员权限
- 不能安装在虚拟环境中 (Vmware, VirtualBox, EC2等)
- 一台机器只能装一个版本DSLS服务器
- 安装新的版本DSLS之前, 需要把旧的版本DSLS服务器卸载掉
- DSLS服务器监听3种端口
 - 4084 : 接受来自DSLS Admin tool的连接
 - **4085 : 客户端使用**
 - 4086 : 冗余服务器端口 (服务器直接通信)

TARGET ID

- 获取默认target ID —— DSLicTarget.exe -t
- Target ID列表 —— DSLicTarget.exe -l
- 不能获得虚拟环境target ID

```
选择 命令提示符
Microsoft Windows [Version 10.0.19044.2251]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JLI20>"C:\Users\JLI20\Desktop\DSLicTarget120\Windows 64-bit\DSLicTarget.exe" -t
DSLicTarget (version 1.20)
Computer Id: NHK-42D218F3ABA14A3C (based on: Intel(R) Ethernet Connection (11) I219-LM)

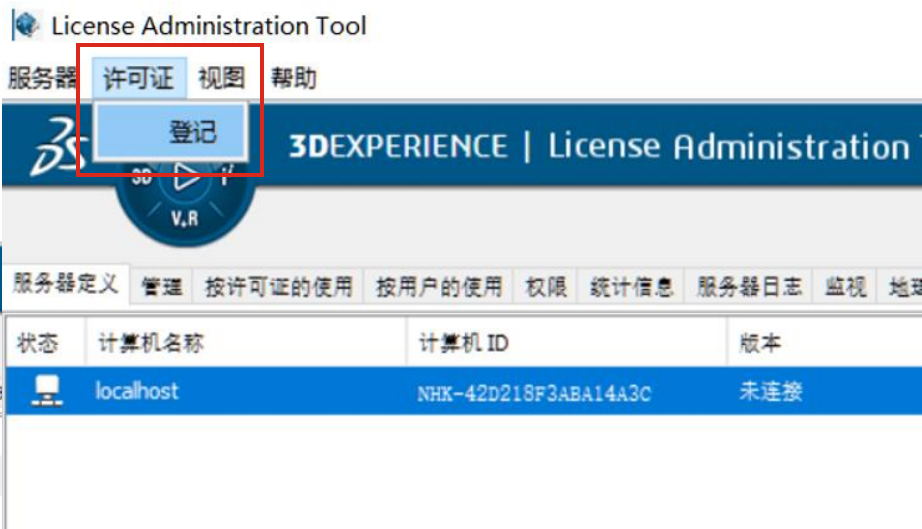
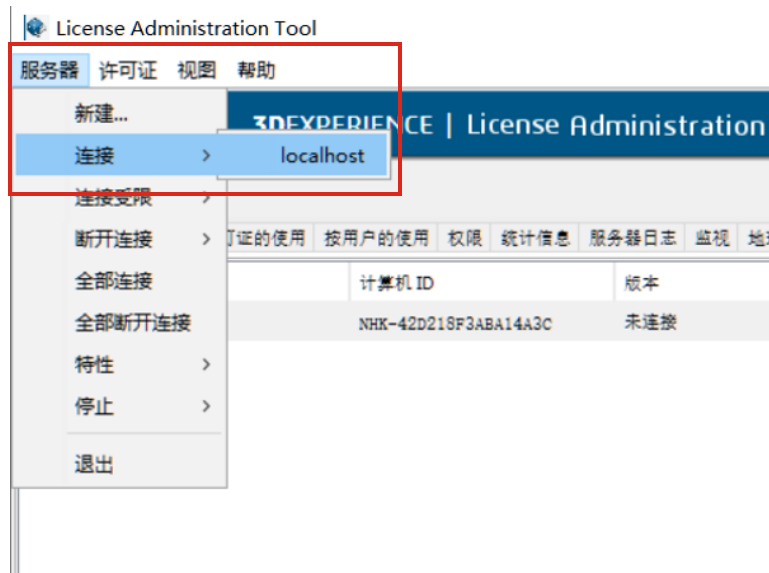
C:\Users\JLI20>"C:\Users\JLI20\Desktop\DSLicTarget120\Windows 64-bit\DSLicTarget.exe" -l
DSLicTarget (version 1.20)
Adapter :
    Computer Id: NHK-42D218F3ABA14A3C
    Description: Intel(R) Ethernet Connection (11) I219-LM
    Type       : Ethernet
    Identifier  : {5FF8975B-D2E0-4A02-A5B0-C8CCDC54EF91}

Adapter :
    Computer Id: BMH-435B145CF3FCBC4B
    Description: Intel(R) Wi-Fi 6 AX201 160MHz
    Type       : Wireless
    Identifier  : {1CDEC540-2C59-40E4-9E4F-D630A106EA79}

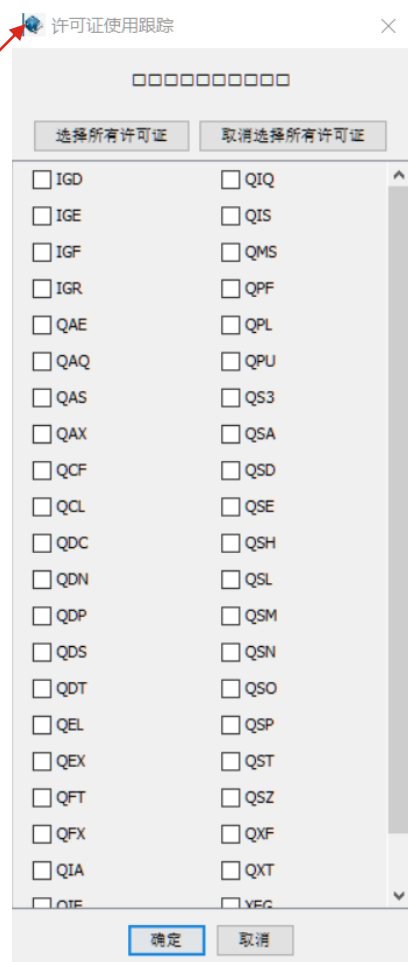
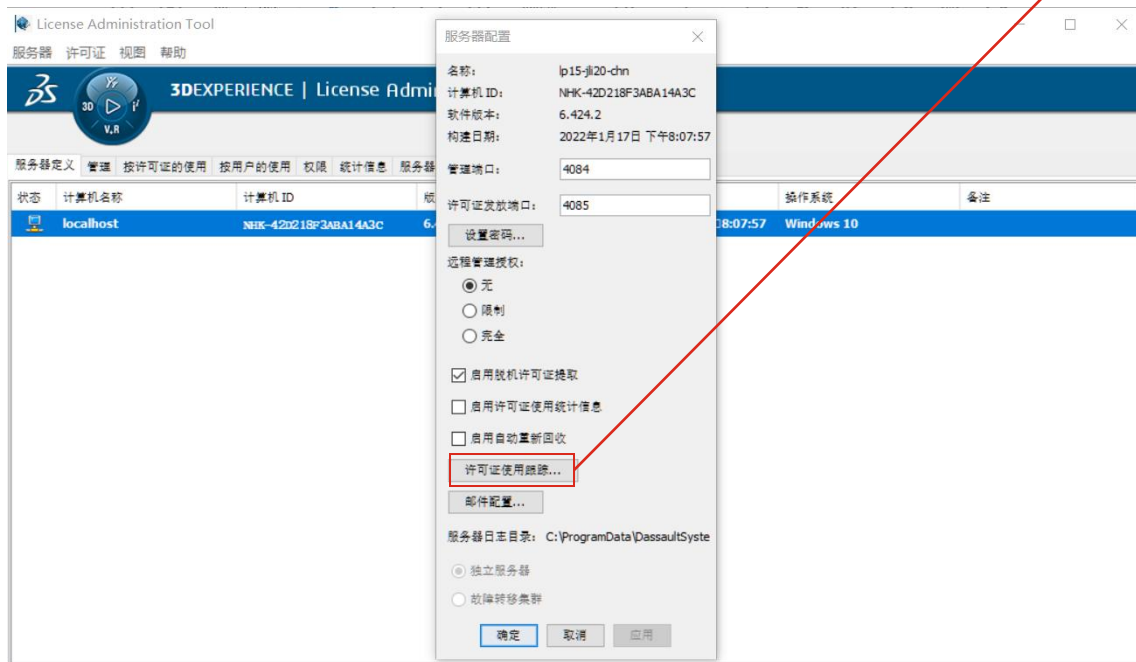
C:\Users\JLI20>
```

服务器配置

- 服务器连接
- Enroll license 文件(.licz)



服务器配置



查看DSLS基本信息

- Abaqus command: abaqus licensing dslsStat -usage

```
d:\temp>abaqus licensing dslsStat -usage
# 1 license servers declared:
Server: lp15-jli20-chn
  Name       : lp15-jli20-chn
  Port       : 4085
  Status     : Running
  Type       : Standalone
  Version    : 6.424.2
  Build date : 1? 17, 2022, 08:07:57 ??
  OS Name    : Windows 10
  OS Version : 10.0
  OS Arch    : amd64
  Computer Id : NHK-42D218F3ABA14A3C

Licenses:
# Feature
```

Feature	Version	Max Release	Model	Type	Number	InUse	Expires	Server Name	Customer ID
IGD	23	21-11?-2023 07:59:00	ConcurrentUser	STD	100	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
IGD	24	21-11?-2023 07:59:00	ConcurrentUser	STD	20	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
IGD	24	21-11?-2023 07:59:00	ConcurrentUser	STD	100	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
IGE	23	21-11?-2023 07:59:00	ConcurrentUser	STD	1	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
IGF	23	21-11?-2023 07:59:00	Token	STD	1000	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
IGF	24	21-11?-2023 07:59:00	Token	STD	1000	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
IGR	23	21-11?-2023 07:59:00	ConcurrentUser	STD	100	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
IGR	24	21-11?-2023 07:59:00	ConcurrentUser	STD	40	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
IGR	24	21-11?-2023 07:59:00	ConcurrentUser	STD	100	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QAE	23	22-9?-2023 07:59:00	Token	STD	20	0	22-9?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QAE	23	21-11?-2023 07:59:00	Token	STD	100	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QAE	24	22-9?-2023 07:59:00	Token	STD	17	0	22-9?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QAE	24	21-11?-2023 07:59:00	Token	STD	100	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QAA	23	21-11?-2023 07:59:00	ConcurrentUser	STD	1	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QAA	24	21-11?-2023 07:59:00	ConcurrentUser	STD	1	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QAS	23	21-11?-2023 07:59:00	ConcurrentUser	STD	1	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QAS	24	21-11?-2023 07:59:00	ConcurrentUser	STD	1	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QAX	23	21-11?-2023 07:59:00	Token	STD	100	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QAX	24	21-11?-2023 07:59:00	Token	STD	100	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QCF	23	21-11?-2023 07:59:00	Token	STD	1000	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QCF	24	21-11?-2023 07:59:00	Token	STD	1000	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QCL	23	22-9?-2023 07:59:00	ConcurrentUser	STD	1	0	22-9?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QCL	23	21-11?-2023 07:59:00	ConcurrentUser	STD	2	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QCL	24	22-9?-2023 07:59:00	ConcurrentUser	STD	1	0	22-9?-2023 07:59:00	lp15-jli20-chn	DSCHN028
QCL	24	21-11?-2023 07:59:00	ConcurrentUser	STD	2	0	21-11?-2023 07:59:00	lp15-jli20-chn	DSCHN028

客户端更新配置

- 指定license服务类型 (DSLS/Flexnet)

Abaqus_installation_dir\2022\win_b64\SMA\site

```
# Installation of Established Products 2022
# Sat May 21 16:28:12 2022
plugin_central_dir="d:\SIMULIA\CAE\plugins\2022"
license_server_type=DSLS
# retrieve licensing configuration from EstablishedProductsConfig.ini
importEnv('licensing.env')
```

Custom_v6.env

```
# licensing environment file for Established Products

from readLicenseConfig import *

# read in configuration from the EstablishedProductsConfig.ini file
config = LicenseConfig("Abaqus", False) # verbose output = True/False

# set Abaqus licensing parameters

# License Server type
value=config.getLicenseServerType()
if value == 'flex':
    license_server_type=FLEXNET
if value == 'dsls':
    license_server_type=DSLS

# FLEX License config
value=config.getFlexLicenseConfig()
if value != '':
    abaquslm_license_file=value

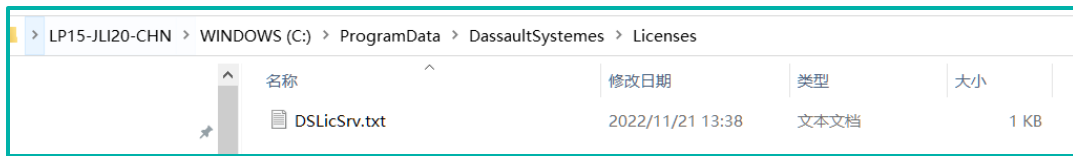
# DSLS License config
value=config.getDslsLicenseConfig()
if value != '':
    dsls_license_config=value.replace('\n', '\\n')
```

licensing.env

客户端更新配置

- Windows

- C:\ProgramData\DassaultSystemes\Licenses
- 创建ANSI文本文件DSLicSrv.txt
- Server_name:port 比如 localhost:4085



- Linux

- /var/DassaultSystemes/Licenses

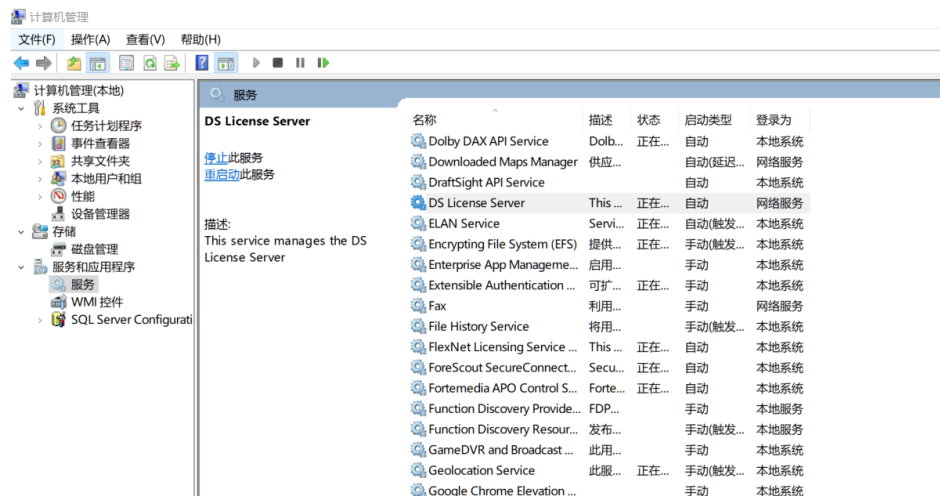
诊断

• 服务停止

- 防火墙
- 计算机休眠时间
- 网卡问题
- 支持的操作系统

• Common Error: 1A000062 , 1A000048

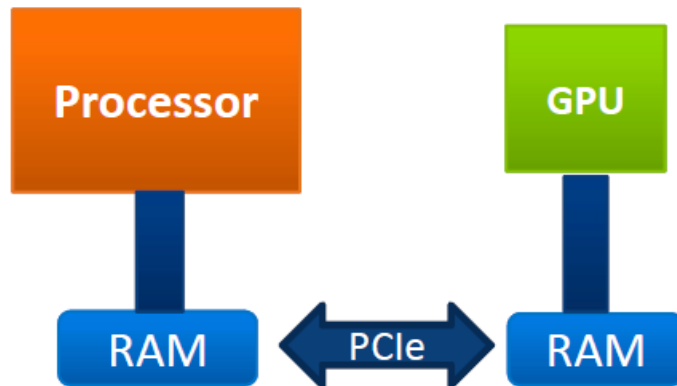
- [01/01] vxn1dsi:4085 KO: Failed to open connection , intermittent checkout failure with DSLS license server
- Bug, 更新版本即可
- Local DSLS licenses can not be checked out when internet connection is not available
- 使用localhost:4085代替hostname:4085



GPU及子程序

GPGPU计算

- GPGPU-General Purpose Graphics Processing Unit
- 使用GPU做通用计算
- CPU-GPU通过PCI总线通信
- 每个GPU都有自己的内存空间



ABAQUS GPGPU计算

- Abaqus以下功能支持GPU计算

- Standard : Direct Sparse Solver
- Abaqus/AMS

- Abaqus支持多GPU多核

- 推荐一个socket一个GPU卡

- Token计算时一个GPU当作一个核

- 所支持的GPU卡参考前述安装准备

- 限制

- 适用于大模型, 实体单元
- 不适用于小模型、shell或结构单元
- 不能用于Explicit计算

Mode name

GPU name

```
C:\Program Files\NVIDIA Corporation\NVSMI>nvidia-smi.exe
Wed Apr 13 22:19:36 2022
```

NVIDIA-SMI 462.59		Driver Version: 462.59		CUDA Version: 11.2	
GPU	Name	TCC/WDDM	Bus-Id	Disp.A	Volatile Uncorr. ECC
Fan	Temp	Perf	Pwr:Usage/Cap	Memory-Usage	GPU-Util Compute M. MIG M.
0	Quadro T1000	WDDM	00000000:01:00:0	On	N/A
N/A	43C	P8	3W / N/A	808MiB / 4096MiB	0% Default N/A

Processes:						
GPU	GI ID	CI ID	PID	Type	Process name	GPU Memory Usage
0	N/A	N/A	2944	C+G	...2txyewy\TextInputHost.exe	N/A
0	N/A	N/A	4284	C+G	...as_DLO\DL0\DL0Clientu.exe	N/A
0	N/A	N/A	5020	C+G	...Office\Office16\lync.exe	N/A
0	N/A	N/A	9656	C+G	...8wekyb3d8bbwe\Cortana.exe	N/A
0	N/A	N/A	12464	C+G	...fice\Office16\OUTLOOK.EXE	N/A
0	N/A	N/A	13868	C+G	...m Files\Zoom\bin\Zoom.exe	N/A
0	N/A	N/A	14428	C+G	...erver\YourPhoneServer.exe	N/A
0	N/A	N/A	14604	C+G	...raXpress\jabra-direct.exe	N/A
0	N/A	N/A	15512	C+G	...ice\Office16\POWERPNT.EXE	N/A
0	N/A	N/A	15744	C+G	...b3d8bbwe\ScreenSketch.exe	N/A
0	N/A	N/A	18024	C+G	..._b64\code\bin\ABQcaeK.exe	N/A
0	N/A	N/A	21640	C+G	Insufficient Permissions	N/A
0	N/A	N/A	22020	C+G	...cw5n1h2txyewy\LockApp.exe	N/A

子程序安装

- Windows
 - 先安装VS C++, 然后安装intel fortran
- Linxu
 - 只需安装intel fortran (操作系统自带GCC)

➤ C:\SIMULIA\Commands\ abaqus.bat

```
@echo off
IF NOT DEFINED PRODUCT_NAME call "C:\Program Files (x86)\IntelSWTools\compilers_and_libraries_2020.0.166\windows\bin\ipsxe-comp-vars.bat" intel64 vs2019
"d:\SIMULIA\Commands\abq2021hf9.bat" %*
```

